

Linux 2FA Security Cheat Sheet

Goal: Protect SSH and physical TTY/HDMI console logins with PAM + Google Authenticator TOTP.

■ **CRITICAL:** Keep your current terminal/session open until a **second terminal successfully logs in with 2FA**. A PAM/SSH typo can lock you out. Keep your emergency scratch codes somewhere secure.

1. Install PAM 2FA Module

```
sudo apt update
sudo apt install libpam-google-authenticator -y
```

2. Generate the User's 2FA Secret

Run as your normal user — NOT with sudo:

```
google-authenticator
```

Prompts: Time-based tokens → **y** → scan QR code → save emergency scratch codes → update file → **y** → disallow token reuse → **y** → default 30-second window → **n** → rate limiting → **y**.

3. Configure SSH PAM

```
sudo nano /etc/pam.d/sshd
```

Find `@include common-auth` and add directly below it:

```
auth required pam_google_authenticator.so
```

Save: **Ctrl+O** → **Enter** → **Ctrl+X**

4. Configure the SSH Daemon

```
sudo nano /etc/ssh/sshd_config
```

Ensure these settings are present/enabled:

```
KbdInteractiveAuthentication yes
UsePAM yes
AuthenticationMethods keyboard-interactive
```

Important: Before restarting SSH, validate the configuration if your system supports it: `sudo sshd -t`. If it reports an error, fix it before restarting.

```
sudo systemctl restart ssh
```

5. Configure Physical TTY / HDMI Login

```
sudo nano /etc/pam.d/login
```

Find `@include common-auth` and add directly below it:

```
auth required pam_google_authenticator.so
```

No service restart is normally required; it applies to subsequent local logins.

6. Lifeline Test — DO NOT CLOSE THE CURRENT SESSION

1	Open a brand-new terminal on your laptop/phone.
2	Connect: <code>ssh redwannabil@<your-pi-ip></code>
3	Enter your normal Linux password.
4	When prompted with "Verification code:", enter the 6-digit TOTP code.
5	Confirm you receive a working shell in the new session.

7. Final Checklist

- Google Authenticator secret generated for the correct Linux user
- QR code scanned successfully
- Emergency scratch codes stored securely
- `/etc/pam.d/sshd` updated
- `/etc/ssh/sshd_config` updated
- sshd configuration tested successfully
- SSH restarted
- `/etc/pam.d/login` updated
- Second SSH session successfully authenticated with 2FA

■ Original terminal kept open until the test succeeded

Recovery principle: If the new SSH session fails, keep the original session open and undo/fix the most recent PAM or SSH change. Do not log out of the working session until access has been verified.